Integrals 50 Practice Questions

Grade 12 | Free

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This free set moves from antiderivatives to definite integrals and reasoning. Attempt questions before reading the companion solutions. All angles are in radians; ln is the natural logarithm. Give exact answers. Indefinite integrals include C and are considered on intervals where their integrands are defined. $\lfloor u \rfloor$ denotes the greatest integer not exceeding u.

Getting started

1. Evaluate $\int (7x^3 - 4x + 9)\,dx$.

2. Evaluate $\int (5\sqrt{x} - 3/x^2)\,dx$, $x > 0$.

3. Find F if $F'(x) = 6x^2 - 5$ and $F(2) = 7$.

4. Evaluate $\int (4\cos 3x - 5\sin 2x)\,dx$.

5. Evaluate $\int (2e^{4x} + 3^x)\,dx$.

6. Evaluate $\int \frac{6x}{3x^2+8}\,dx$.

7. Evaluate $\int x(2x^2 + 5)^4\,dx$.

8. Evaluate $\int \frac{\cos x}{4+\sin x}\,dx$.

9. Evaluate $\int \frac{dx}{9+4x^2}$.

10. Evaluate $\int \frac{dx}{\sqrt{25-9x^2}}$, $|x| < 5/3$.

Choosing and combining methods

11. Evaluate $\int \frac{2x+7}{x^2+7x+15}\,dx$.

12. Evaluate $\int \frac{dx}{x^2+6x+13}$.

13. Evaluate $\int \frac{dx}{(x+2)(x+5)}$.

14. Evaluate $\int \frac{3x+8}{x^2+5x+6}\,dx$.

15. Evaluate $\int \frac{x^2+2}{x^2+4}\,dx$.

16. Evaluate $\int xe^{3x}\,dx$.

17. Evaluate $\int x\sin 4x\,dx$.

18. Evaluate $\int \ln(2x)\,dx$, $x > 0$.

19. Evaluate $\int x^2 \ln x\,dx$, $x > 0$.

20. Evaluate $\int \sin^2(3x)\,dx$.

21. Evaluate $\int \sin^3 x \cos x \, dx$.

22. Evaluate $\int \sec^2 x \tan^2 x \, dx$ on an interval without poles.

23. Evaluate $\int \frac{dx}{\sqrt{x}+2}$, $x > 0$.

24. Evaluate $\int \frac{dx}{x(3+\ln x)^2}$, $x > 0$, $x \neq e^{-3}$.

25. Evaluate $\int \frac{2x+1}{\sqrt{x^2+x+6}} \, dx$.

Definite integrals

26. Evaluate $\int_{-1}^{2} (3x^2 + 2) \, dx$.

27. Evaluate $\int_{0}^{2} \frac{x}{x^2+5} \, dx$.

28. Evaluate $\int_{0}^{\pi/6} \sin 3x \, dx$.

29. Evaluate $\int_{1}^{3} \frac{dx}{(x+1)^2}$.

30. Evaluate $\int_{-2}^{2} \frac{x^5}{3+x^2} \, dx$.

31. Evaluate $\int_{-2}^{2} (x^4 + 3x^2) \, dx$.

32. Evaluate $\int_{0}^{5} \frac{(x+1)^2}{(x+1)^2+(6-x)^2} \, dx$.

33. Evaluate $\int_{0}^{3} |2x - 1| \, dx$.

34. For $f(x) = x + 2$ on $[0, 1]$ and $f(x) = 4 - x$ on $(1, 3]$, find $\int_{0}^{3} f(x) \, dx$.

35. Evaluate $\int_{0}^{5/2} \lfloor 2x \rfloor \, dx$.

36. Evaluate $\int_{0}^{1} x \ln(1 + x^2) \, dx$.

37. Evaluate $\int_{0}^{2} xe^{2x} \, dx$.

38. Evaluate $\int_{0}^{\pi/2} \frac{\sin x \cos x}{2+\sin^2 x} \, dx$.

Further reasoning

39. Find $G'(x)$ for $G(x) = \int_{1}^{x^2} (1 + t^4) \, dt$.

40. Find $H'(x)$ for $H(x) = \int_{x}^{3x} \frac{dt}{1+t^2}$.

41. Evaluate $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \frac{1}{3+r/n}$.

42. Evaluate $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{r^2}{n^3+nr^2}$.

43. Find c such that $\int_{0}^{3} (x - c)(x + 1) \, dx = 0$.

44. Find the minimum of $J(a) = \int_{-1}^{2} (x-a)^2\,dx$ over real a.

45. Evaluate $\int_0^1 \frac{x^3}{1+x^2}\,dx$.

46. Evaluate $\int_0^{\pi/2} \frac{dx}{9\cos^2 x + 4\sin^2 x}$.

47. Let $I_n = \int_0^1 x^n (1-x)^3\,dx$ for integers $n \geq 0$. Find I_n.

48. A continuous function satisfies $f(x) = 3 + \int_0^x 2t f(t)\,dt$. Find f and $\int_0^1 x f(x)\,dx$.

49. Without finding an antiderivative, show $1/6 < \int_0^1 \frac{dx}{5+x^4} < 1/5$.

50. Let $F(x) = \int_0^x (t-1)(t-3)\,dt$. Find its stationary points and classify them.